

Project Snapshot

This portfolio project models a BI role at a fictional medium sized manufacturer. I worked through seven CRM and ERP style source files from raw data through cleanup, QA, SQL analysis, Power BI modeling, DAX, report design, and final validation.

51,038 Cleaned rows	\$310.32M Won sales	\$283.73M Shipped revenue	\$15.13M Current backlog
\$33.24M Open pipeline	46.27% Shipped margin	91.93% On-time	13 Report pages

What I Built

- A clean analytical layer across Accounts, Opportunities, Orders, Order Lines, Returns, Warranty, and Product Reference.
- A pandas QA script that checks keys, relationships, calculations, shipped and backlog logic, order header reconciliation, and the handoff from Won opportunities to ERP orders.
- A SQLite database and SQL analysis covering the main sales, operations, margin, quality, and customer questions.
- A Power BI model with shared dimensions and separate fact tables. The final report has 12 main pages plus Account 360 drillthrough.

Technical Validation

Check	Final Result
Order line calculations	0 mismatches in margin, total, margin %, shipped revenue, shipped quantity, backlog value, and backlog quantity
Header to line reconciliation	0 mismatches in Value, Freight, Tax, Total, or COGS
Core relationships	0 invalid cleaned relationships across orders, accounts, products, returns, and warranty
Exceptions kept visible	78 order lines and 23 orders flagged for review. 69 unresolved opportunity Account IDs. 26 Won opportunities without a standard ERP order

Strongest Business Findings

- Won sales reached \$75.38M through September 2026, already above the \$73.19M full year 2025 result.
- Of the \$33.24M in open pipeline, \$19.55M was stale, or 58.8%.
- Office Chairs produced about 65.1% of shipped revenue and 74.6% of current backlog.
- Competitor Selected and Price accounted for about 46.8% of lost opportunity value.
- The top 10 accounts were about 4.4% of shipped revenue. Repeat business among shipped accounts was about 87.0%.
- Standard order on-time performance was 91.93%.

Final Reporting Decisions

- Current Open Pipeline and Current Backlog are current snapshot measures, so they ignore the Date slicer.
- The Date table extends through January 2027 so future projected opportunities are retained in the model.
- The Delays & Operations page is a current work queue for open orders with an Adjusted Promised Ship Date. Closed and Cancelled orders are excluded.
- Weighted Account Ranking uses Won Value at 30%, Average Margin at 20%, Won Orders at 15%, Pipeline Value at 15%, Return Rate at 10%, and Warranty Rate at 10%. Lower return and warranty rates score better.